

Course: B. Tech.

Semester: III

Branch: Electrical Engineering / Electrical and Electronics Engineering / Electrical and Power / Electronics and Power

Subject Code & Name: BTES305 Engineering Material Science

Max Marks: 60

Date:09/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Attempt any TWO of the following.		12
A) Distinguish between Electronic conduction & Ionic conduction.	CO1	6
B) Explain classification of Conducting material? Write its properties in details.	CO1	6
C) Explain Quantum free electron theory.	CO1	6
Q.2 Attempt any TWO of the following.		12
A) Explain the behavior of Dielectrics in static and alternating field.	CO2	6
B) Derive Clausius – Mosotti relation.	CO2	6
C) Distinguish between Piezoelectric, Pyroelectric and Ferroelectric materials? Write applications of Dielectric material in details.	CO2	6
Q. 3 Attempt any TWO of the following.		12
A) Explain Integration techniques? Write note on LSI & VLSI circuits.	CO3	6
B) Explain Intrinsic and Extrinsic semiconductor with appropriate diagram.	CO3	6
C) Explain Plain carbon steel, High speed steel and Copper alloy? Write its applications.	CO3	6
Q.4 Attempt any TWO of the following.		12
A) Explain Soft and Hard magnetic materials.	CO4	6
B) Explain Special purpose materials and its magnetic properties.	CO4	6
C) Explain Spontaneous magnetization, Magnetostriction and Diamagnetism in magnetic materials.	CO4	6

Q. 5	Attempt any TWO of the following.		12
A)	What is Structural materials? Explain it in detail.	CO5	6
B)	Explain Galvanization and Impregnation of material? Write its advantages and applications.	CO5	6
C)	What is Refractory materials? Write its classification and applications in details.	CO5	6

***** End *****